CS4095-Embedded Control of Robotics

Assignment # 03, Date: 1/10/2025

Marks = 10 Last.Date: 15/10/2025

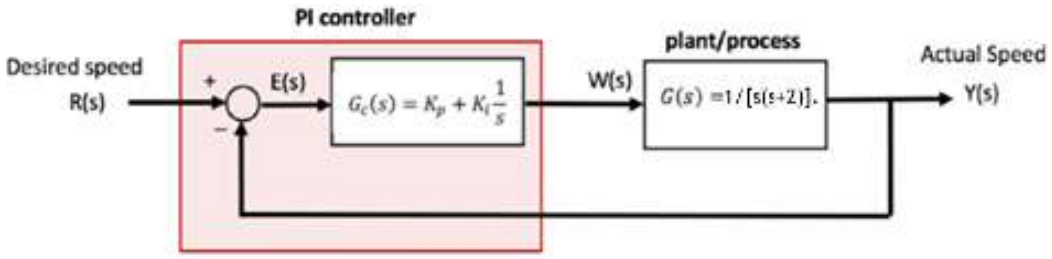
[CLO 3: Design a PID Control system given the specification. (Cognitive, C3)]

A plant has a Transfer Function, G(s) = 1/[s(s+2)].

1. Design a PI controller for feedback that meets the following performance requirements:
    - Percentage overshoot should be less than 4.5%
    - 2% settling time should be less than 2.5 sec

Provide all the necessary steps, calculations, and diagrams to support your design.

$$R(t) = K_P \cdot e(t) + Q_I \cdot \int_0^t e(t)dt$$



2. Using the Python control package, implement this closed-loop system in Python. Plot the step response, PLZ map, and Bode Plot of this system.
    - From the Step plot, write the rise time, overshoot, settling time, and Steady state error.
    - From the PLZ map, write about the poles & zeros of the closed-loop system
    - From the Bode Plot, write about Phase Margin, Gain Margin, Phase shift, and magnitude.

3. Comments on the stability of the System.